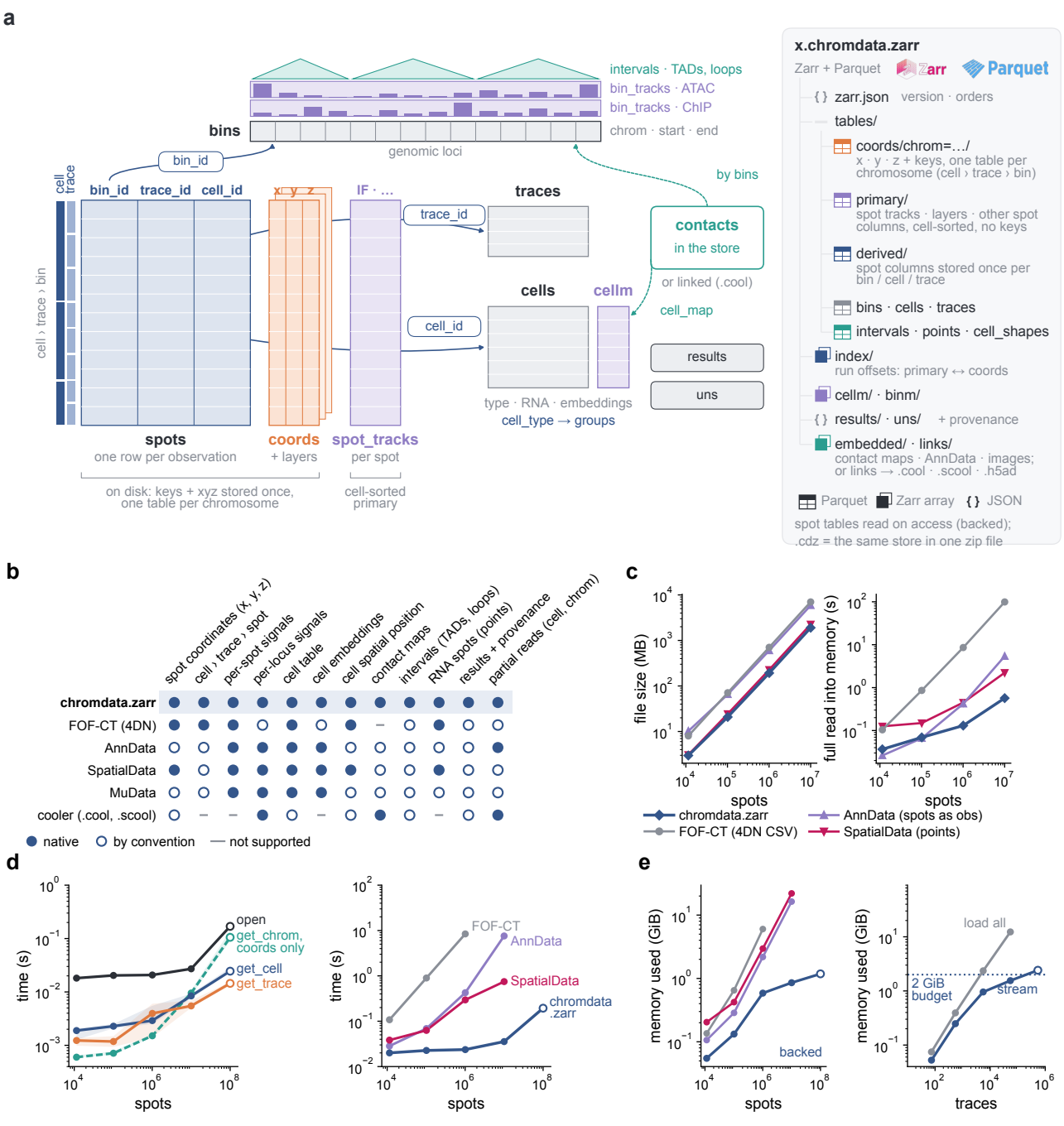

a
intervals · TADs, loops
bin_tracks · ATAC
bin_tracks · ChIP
bins
chrom · start · end
genomic loci
bin_id
cell
trace
bin_id trace_id cell_id
x y z
IF · …
traces
trace_id
by bins
contacts
in the store
or linked (.cool)
cells
cellm
cell_map
cell_id
results
uns
cell › trace › bin
type · RNA · embeddings
cell_type → groups
spots
one row per observation
coords
+ layers
spot_tracks
per spot
on disk: keys + xyz stored once,
one table per chromosome
cell-sorted
primary
x.chromdata.zarr
Zarr + Parquet
zarr
Parquet
zarr.json
version · orders
tables/
coords/chrom=…/
x · y · z + keys, one table per
chromosome (cell › trace › bin)
primary/
spot tracks · layers · other spot
columns, cell-sorted, no keys
derived/
spot columns stored once per
bin / cell / trace
bins · cells · traces
intervals · points · cell_shapes
index/
run offsets: primary ↔ coords
cellm/ · binm/
results/ · uns/
+ provenance
embedded/ · links/
contact maps · AnnData · images;
or links → .cool · .scool · .h5ad
Parquet
Zarr array
JSON
spot tables read on access (backed);
.cdz = the same store in one zip file
b
spot coordinates (x, y, z)
cell › trace › spot
per-spot signals
per-locus signals
cell table
cell embeddings
cell spatial position
contact maps
intervals (TADs, loops)
RNA spots (points)
results + provenance
partial reads (cell, chrom)
chromdata.zarr
FOF-CT (4DN)
AnnData
SpatialData
MuData
cooler (.cool, .scool)
native
by convention
not supported
c
file size (MB)
spots
full read into memory (s)
spots
chromdata.zarr
FOF-CT (4DN CSV)
AnnData (spots as obs)
SpatialData (points)
d
time (s)
spots
open
get_chrom,
coords only
get_cell
get_trace
time (s)
spots
FOF-CT
AnnData
SpatialData
chromdata
.zarr
e
memory used (GiB)
spots
backed
memory used (GiB)
traces
load all
2 GiB
budget
stream